CMPG 325 Computer Networks

Implementation and Testing Report

Milestone 2 evidence and final submission

Dikgosi History & Archives Research Centre (DHARC), Mafikeng

Item	Detail
Student	Tebello Qekisi
Student number	40939170
Project ID	CMPG325-2026-112
Client ID	CLI-112
Assigned IP block	10.42.0.0/16
Assigned feature	Internal HTTP web server in VLAN 40, reachable only from VLAN 10 and VLAN 20
Software	Cisco Packet Tracer 9.0.1
Date	2 October 2026

1. Overview and outcome

This report documents the Packet Tracer implementation of the DHARC campus network designed in Milestone 1, the assigned networking feature, and the evidence that the network works. The network is a collapsed core: one edge router (RTR-EDGE, Cisco 4321), one multilayer core switch (CORE-L3SW) and one Reading Room access switch (ACC-READSW, Cisco 2960). Five VLANs separate Administration, Archives/Research, the public Reading Room, Servers and Management. A two-link EtherChannel joins the core and the access switch.

Before testing, the implementation was audited against the design. The audit found configuration and cabling faults that stopped parts of the network working; they are listed in Section 4 and were corrected before the evidence in Section 6 was captured. All screenshots in this report were taken from the corrected network.

Outcome: all 21 planned tests produced the expected behaviour (Section 5). For two tests, the expected result was refined after observing what Packet Tracer really does: the EtherChannel failover loses about 6 pings rather than none (T7), and Telnet is closed immediately instead of showing a "refused" message (T20). Both are described honestly in the results.

2. Network as implemented

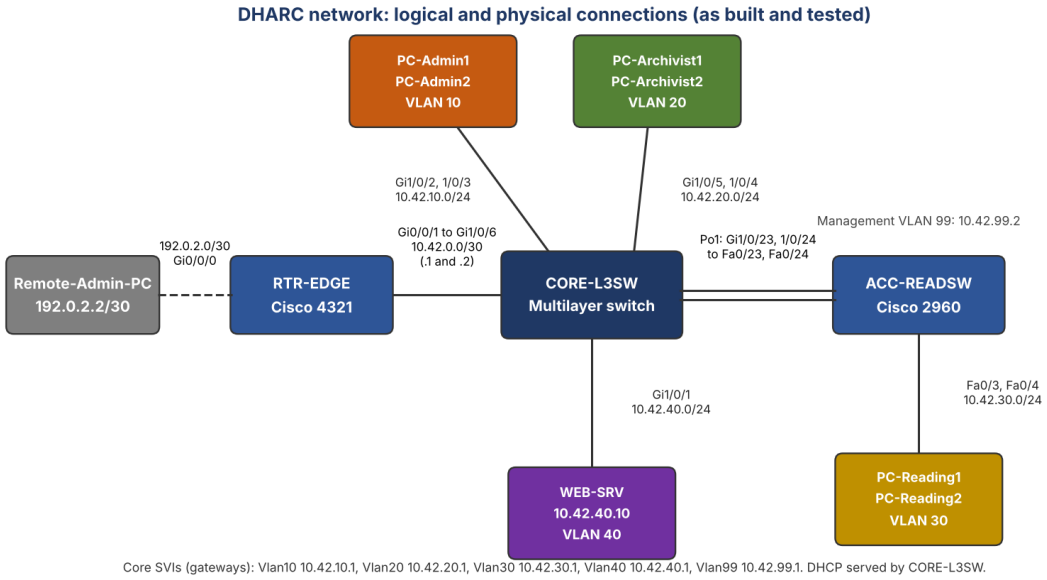


Figure 1: Network topology as built and tested, with ports, subnets and VLANs.

2.1 Devices

Device	Model	Role
RTR-EDGE	Cisco 4321 (ISR4321)	Edge router: link to the off-site administrator (192.0.2.0/30) and to the core
CORE-L3SW	Cisco multilayer switch	Inter-VLAN routing, DHCP server, ACL enforcement, Admin, Archives and Server access
ACC-READSW	Cisco 2960-24TT	Reading Room access switch
WEB-SRV	Server-PT	Internal HTTP web server, VLAN 40
PC-Admin1, PC-Admin2	PC-PT	Administration workstations, VLAN 10
PC-Archivist1, PC-Archivist2	PC-PT	Archives and research workstations, VLAN 20
PC-Reading1, PC-Reading2	PC-PT	Reading Room public terminals, VLAN 30
Remote-Admin-PC	PC-PT	Off-site administrator (192.0.2.2/30)

2.2 Cabling

From	Port	To	Port
Remote-Admin-PC	Fa0	RTR-EDGE	Gi0/0/0
RTR-EDGE	Gi0/0/1	CORE-L3SW	Gi1/0/6
CORE-L3SW	Gi1/0/23	ACC-READSW	Fa0/23 (Po1)
CORE-L3SW	Gi1/0/24	ACC-READSW	Fa0/24 (Po1)
CORE-L3SW	Gi1/0/1	WEB-SRV	Fa0
CORE-L3SW	Gi1/0/2	PC-Admin1	Fa0
CORE-L3SW	Gi1/0/3	PC-Admin2	Fa0
CORE-L3SW	Gi1/0/5	PC-Archivist1	Fa0
CORE-L3SW	Gi1/0/4	PC-Archivist2	Fa0
ACC-READSW	Fa0/3	PC-Reading1	Fa0
ACC-READSW	Fa0/4	PC-Reading2	Fa0

2.3 VLANs and IP addressing

VLAN	Name	Subnet	Gateway (core SVI)	Addressing
10	Administration	10.42.10.0/24	10.42.10.1	DHCP, .1 to .9 excluded
20	Archives-Research	10.42.20.0/24	10.42.20.1	DHCP, .1 to .9 excluded
30	Reading-Room	10.42.30.0/24	10.42.30.1	DHCP, .1 to .9 excluded
40	Servers	10.42.40.0/24	10.42.40.1	Static: WEB-SRV 10.42.40.10
99	Management	10.42.99.0/24	10.42.99.1	Static: ACC-READSW 10.42.99.2

Link	Subnet	Addresses
RTR-EDGE to CORE-L3SW (transit)	10.42.0.0/30	RTR-EDGE Gi0/0/1 10.42.0.1; CORE-L3SW Gi1/0/6 10.42.0.2
RTR-EDGE to off-site administrator	192.0.2.0/30	RTR-EDGE Gi0/0/0 192.0.2.1; Remote-Admin-PC 192.0.2.2

The transit subnet is taken from the assigned 10.42.0.0/16 block and does not overlap any VLAN. The 192.0.2.0/30 range is the documentation range and stands in for the off-site administrator's connection.

2.4 Core services

- **Layer 2:** VLANs 10, 20, 30, 40 and 99 exist on CORE-L3SW; VLANs 30 and 99 exist on ACC-READSW. Po1 is a static EtherChannel (channel-group 1 mode on) over Gi1/0/23 and Gi1/0/24 on the core and Fa0/23 and Fa0/24 on the access switch, configured as a trunk. Either cable can fail without isolating the Reading Room (the no-downtime requirement in Section 8 of the brief).
- **Layer 3:** inter-VLAN routing runs on CORE-L3SW through the SVIs above (ip routing). The core has a default route to RTR-EDGE (10.42.0.1) and the router has a route for 10.42.0.0/16 via the core (10.42.0.2). Static routing is sufficient for one edge router and one core.
- **DHCP:** CORE-L3SW serves ADMIN-POOL, ARCHIVE-POOL and READING-POOL, each with its own default router. The first nine addresses of each subnet are excluded so gateways and future fixed devices keep predictable addresses. Servers and management devices use static addresses.
- **Remote management (Change Request CR9):** all three devices accept SSH version 2 only (transport input ssh, login local). RTR-EDGE accepts sessions only from the off-site

administrator (access-list 10, host 192.0.2.2). CORE-L3SW and ACC-READSW accept sessions only from ACL MGMT-SSH: the off-site administrator, the Administration VLAN and the Management VLAN.

3. Assigned feature: restricted internal web server

Requirement: the internal HTTP server in VLAN 40 must be reachable only from VLAN 10 (Administration) and VLAN 20 (Archives/Research).

3.1 Implementation

WEB-SRV (10.42.40.10) has its HTTP service enabled. The restriction is enforced on CORE-L3SW by the extended ACL WEB-ACCESS, applied outbound on interface Vlan40. Every inter-VLAN packet towards the servers leaves the core through Vlan40, so one ACL covers all other sources, including the Reading Room, the management VLAN and the off-site administrator.

```
ip access-list extended WEB-ACCESS
 permit tcp 10.42.10.0 0.0.0.255 host 10.42.40.10 eq www
 permit tcp 10.42.20.0 0.0.0.255 host 10.42.40.10 eq www
 permit icmp 10.42.10.0 0.0.0.255 10.42.40.0 0.0.0.255
 permit icmp 10.42.20.0 0.0.0.255 10.42.40.0 0.0.0.255
 deny ip any 10.42.40.0 0.0.0.255
 permit ip any any
!
interface Vlan40
 ip address 10.42.40.1 255.255.255.0
 ip access-group WEB-ACCESS out
```

3.2 Supporting control for the Reading Room

The Reading Room is public-facing, so ACL READING-ISOLATE is applied inbound on Vlan30 and denies traffic from VLAN 30 to VLANs 10, 20, 40 and 99. Reading Room PCs still reach their own gateway. This gives two independent layers keeping the public terminals away from the server.

```
ip access-list extended READING-ISOLATE
 deny ip 10.42.30.0 0.0.0.255 10.42.10.0 0.0.0.255
 deny ip 10.42.30.0 0.0.0.255 10.42.20.0 0.0.0.255
 deny ip 10.42.30.0 0.0.0.255 10.42.40.0 0.0.0.255
 deny ip 10.42.30.0 0.0.0.255 10.42.99.0 0.0.0.255
 permit ip any any
!
interface Vlan30
 ip access-group READING-ISOLATE in
```

3.3 Design decisions

- **Outbound on Vlan40:** one ACL protects the whole server VLAN, and any new user VLAN is blocked from it by default without editing that VLAN's interface.
- **Ordering:** specific permits first, then an explicit deny for the server subnet, then permit ip any any so that no other traffic leaving through Vlan40 is affected. Without the final permit the implicit deny would drop every other packet leaving that interface.
- **ICMP from VLANs 10 and 20:** lets staff test the server with ping. Only port 80 is open for applications; other protocols from those VLANs, such as SSH or FTP, are blocked.
- **Host-specific permit:** web access is permitted to host 10.42.40.10 only, not the whole VLAN, so a future server in VLAN 40 is not exposed by accident.
- **Two layers for the Reading Room:** READING-ISOLATE stops Reading Room traffic first, so its blocked packets are counted there; WEB-ACCESS is still the control that protects the server from every other source (shown in T17).

4. Faults found in the audit and corrections made

No.	Fault found	Effect	Correction
1	Reading Room PCs were cabled to Fa0/3 and Fa0/4 on ACC-READSW, but only Fa0/1 and Fa0/2 were set to VLAN 30.	PCs stayed in VLAN 1 and could not obtain DHCP addresses (Figure 3: no VLAN 30 leases).	Fa0/3 and Fa0/4 set as access ports in VLAN 30.
2	EtherChannel Po1 on ACC-READSW was configured on Gi0/1 and Gi0/2, but the two core cables are connected to Fa0/23 and Fa0/24.	No working bundle on the access side (Figure 2: Po1 members down).	Po1 members on ACC-READSW changed to Fa0/23 and Fa0/24, where the cables are.
3	RTR-EDGE Gi0/0/1 had no IP address and neither device had a route to the other.	No path between the campus and the edge router or the off-site administrator.	Transit link 10.42.0.0/30 added; core Gi1/0/6 made a routed port; default route on the core and a 10.42.0.0/16 route on the router.
4	CORE-L3SW and ACC-READSW had no enable secret, and user tt had privilege 1.	An SSH login could not reach privileged mode.	User tt set to privilege 15 on all three devices, with a local password.
5	Only the router restricted SSH by ACL.	Switches accepted SSH from any VLAN, against Change Request CR9.	ACL MGMT-SSH added to the core and access switch vty lines.
6	No device had a saved startup configuration.	A reload would have erased all configuration.	Configuration saved on all three devices (T21).
7	RSA keys were missing on all three devices.	SSH cannot start without a key pair.	Keys generated on each device (Figures 4 to 6).

```
ACC-READSW>show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/23      auto         n-802.1q       trunking      1
Fa0/24      auto         n-802.1q       trunking      1

Port        Vlans allowed on trunk
Fa0/23      1-1005
Fa0/24      1-1005

Port        Vlans allowed and active in management domain
Fa0/23      1,30,99
Fa0/24      1,30,99

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/23      1,30,99
Fa0/24      1,30,99

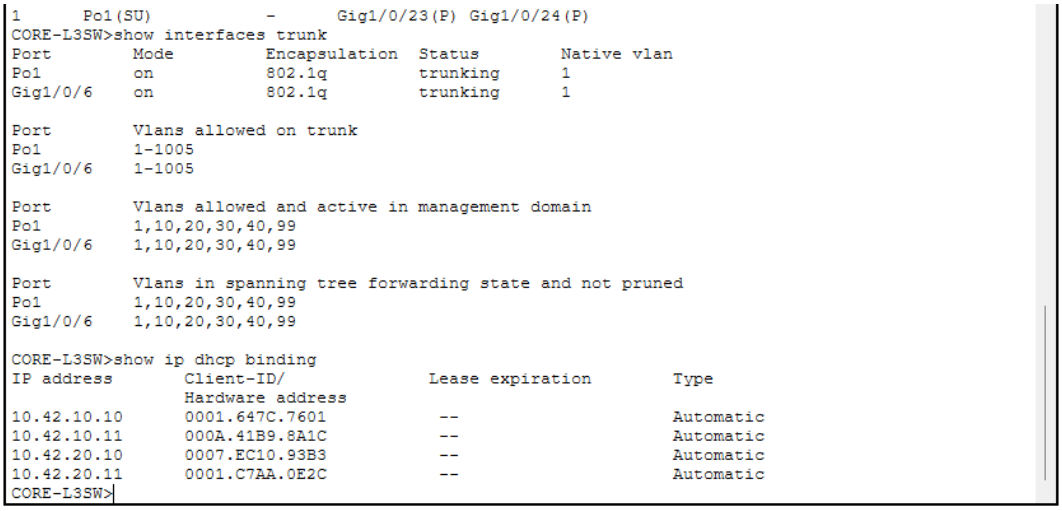
ACC-READSW>show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------

1      Po1(SD)           -      Gig0/1(D) Gig0/2(D)
ACC-READSW>show running-config interface fa0/3
```

Figure 2: Before correction, ACC-READSW: the trunk runs on Fa0/23 and Fa0/24, but Po1 lists Gi0/1 and Gi0/2 (both down) as its members.



```
1      Po1(SU)           -      Gig1/0/23(P) Gig1/0/24(P)
CORE-L3SW>show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Po1         on           802.1q         trunking      1
Gig1/0/6    on           802.1q         trunking      1

Port        Vlans allowed on trunk
Po1         1-1005
Gig1/0/6    1-1005

Port        Vlans allowed and active in management domain
Po1         1,10,20,30,40,99
Gig1/0/6    1,10,20,30,40,99

Port        Vlans in spanning tree forwarding state and not pruned
Po1         1,10,20,30,40,99
Gig1/0/6    1,10,20,30,40,99

CORE-L3SW>show ip dhcp binding
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
10.42.10.10      0001.647C.7601           --                     Automatic
10.42.10.11      000A.41B9.8A1C           --                     Automatic
10.42.20.10      0007.EC10.93B3           --                     Automatic
10.42.20.11      0001.C7AA.0E2C           --                     Automatic
CORE-L3SW>
```

Figure 3: Before correction, CORE-L3SW: Po1 is up, and DHCP leases exist for VLAN 10 and VLAN 20 only; none for VLAN 30 (Reading Room).

5. Test results

Each test was run in Packet Tracer on the corrected network. The Result column states what was observed, and the Evidence column points to the screenshots in Section 6. Every test produced the expected behaviour.

ID	Test	Result observed	Status	Evidence
T1	DHCP, Administration	PC-Admin1 10.42.10.10 and PC-Admin2 10.42.10.11, mask 255.255.255.0, gateway 10.42.10.1. "DHCP request successful."	Pass	Fig. 7, 8
T2	DHCP, Archives	PC-Archivist1 10.42.20.10 and PC-Archivist2 10.42.20.11, gateway 10.42.20.1. "DHCP request successful."	Pass	Fig. 9, 10
T3	DHCP, Reading Room	PC-Reading1 10.42.30.11 and PC-Reading2 10.42.30.10, gateway 10.42.30.1. "DHCP request successful."	Pass	Fig. 11, 12
T4	Gateway reachability	All six PCs ping their own gateway: 4 sent, 4 received, 0% loss.	Pass	Fig. 14 to 19
T5	Inter-VLAN routing	PC-Admin1 to PC-Archivist1 (10.42.20.10): 4 of 4 replies, TTL 127 (routed through the core).	Pass	Fig. 20
T6	EtherChannel and VLANs	Core: Po1(SU), Gi1/0/23(P), Gi1/0/24(P). Access switch: Po1(SU), Fa0/23(P), Fa0/24(P). VLANs 10, 20, 30, 40, 99 present; Fa0/1 to Fa0/4 in VLAN 30.	Pass	Fig. 21, 22
T7	EtherChannel failover	With one cable removed, a continuous ping from PC-Reading1 to its gateway lost 6 packets, then recovered without intervention. Core showed Po1(SU) with Gi1/0/23(P) and Gi1/0/24(D). After reconnecting, both ports (P).	Pass	Fig. 23 to 25
T8	Feature: Administration allowed	PC-Admin1 loads the web page at http://10.42.40.10.	Pass	Fig. 26
T9	Feature: Archives allowed	PC-Archivist1 loads the web page at http://10.42.40.10.	Pass	Fig. 27
T10	Feature: Reading Room blocked (HTTP)	PC-Reading1 browser shows "Request Timeout".	Pass	Fig. 28
T11	Feature: Reading Room blocked (ping)	PC-Reading1 ping 10.42.40.10: 0 of 4 received, "Destination host unreachable" from 10.42.30.1. Its own gateway still replies.	Pass	Fig. 29

ID	Test	Result observed	Status	Evidence
T12	Feature: ICMP from Administration	PC-Admin1 ping 10.42.40.10: 4 of 4 replies.	Pass	Fig. 30
T13	ACL counters	WEB-ACCESS: HTTP permits for VLAN 10 and VLAN 20 each 5 matches; ICMP permit for VLAN 10: 8; deny for VLAN 40: 21. READING-ISOLATE: denies towards VLAN 10: 4; VLAN 40: 16; VLAN 99: 16.	Pass	Fig. 31
T14	Reading Room isolation	PC-Reading1 to PC-Admin1 (10.42.10.10) and to 10.42.99.1: 0 of 4 received, "Destination host unreachable" from 10.42.30.1. Own gateway still reachable.	Pass	Fig. 32
T15	Router to core link	RTR-EDGE ping 10.42.0.2: 5 of 5. CORE-L3SW ping 10.42.0.1: 5 of 5. RTR-EDGE ping 10.42.10.1: 5 of 5.	Pass	Fig. 33 to 35
T16	Routing from the off-site PC	Remote-Admin-PC ping 192.0.2.1 and 10.42.10.1: 4 of 4 replies each.	Pass	Fig. 36
T17	Server shielded from other sources	Remote-Admin-PC ping 10.42.40.10: "Destination host unreachable" from 10.42.0.2 (0 of 4). RTR-EDGE ping 10.42.40.10: UUUUU (0 of 5).	Pass	Fig. 35, 36
T18	SSH from the off-site administrator	SSH as user tt from Remote-Admin-PC reached RTR-EDGE (192.0.2.1), CORE-L3SW (10.42.99.1) and ACC-READSW (10.42.99.2) at the privileged prompt.	Pass	Fig. 4 to 6, 37
T19	SSH blocked from other sources	PC-Reading1 to 192.0.2.1: connection refused; to 10.42.99.1: timed out. PC-Admin1 to 192.0.2.1: connection refused. CORE-L3SW to 192.0.2.1: connection refused. PC-Admin1 to the core is allowed by design (MGMT-SSH).	Pass	Fig. 37 to 39
T20	Telnet not usable	telnet 192.0.2.1: the connection opens, then the router closes it at once with no login prompt (transport input ssh).	Pass	Fig. 40
T21	Configuration saved	write memory returned [OK] on all three devices, and show startup-config shows the saved configuration.	Pass	Fig. 41 to 43

6. Evidence

6.1 SSH key setup (prerequisite for T18 to T20)

```
RTR-EDGE>enable
RTR-EDGE#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
RTR-EDGE(config)#crypto key generate rsa
The name for the keys will be: RTR-EDGE.dharc.local
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

RTR-EDGE(config)#end
*Mar 1 4:7:46.506: %SSH-5-ENABLED: SSH 2 has been enabled
RTR-EDGE#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
RTR-EDGE#
```

Figure 4: RTR-EDGE: RSA key pair generated (1024 bits), SSH 2 enabled, configuration saved.

```
CORE-L3SW>enable
CORE-L3SW#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
CORE-L3SW(config)#crypto key generate rsa
The name for the keys will be: CORE-L3SW.dharc.local
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

CORE-L3SW(config)#end
*Mar 1 3:46:5.766: %SSH-5-ENABLED: SSH 2 has been enabled
CORE-L3SW#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
CORE-L3SW#
```

Figure 5: CORE-L3SW: RSA key pair generated (1024 bits), SSH 2 enabled, configuration saved.

```
ACC-READSW>enable
ACC-READSW#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
ACC-READSW(config)#crypto key generate rsa
The name for the keys will be: ACC-READSW.dharc.local
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

ACC-READSW(config)#end
*Mar 1 4:8:56.855: %SSH-5-ENABLED: SSH 2 has been enabled
ACC-READSW#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
ACC-READSW#
```

Figure 6: ACC-READSW: RSA key pair generated (1024 bits), SSH 2 enabled, configuration saved.

6.2 Addressing (T1 to T3)

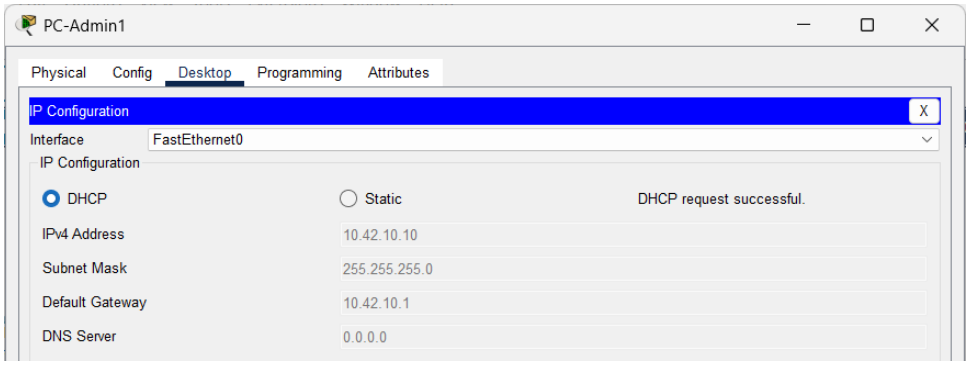


Figure 7: T1: PC-Admin1 receives 10.42.10.10, gateway 10.42.10.1.

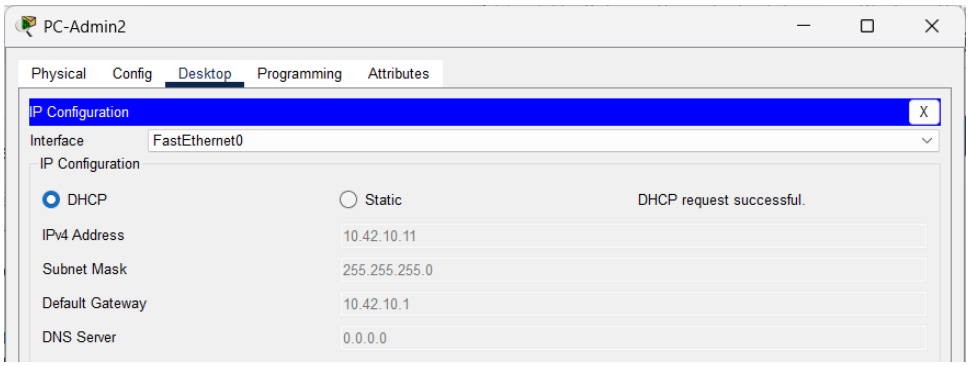


Figure 8: T1: PC-Admin2 receives 10.42.10.11, gateway 10.42.10.1.

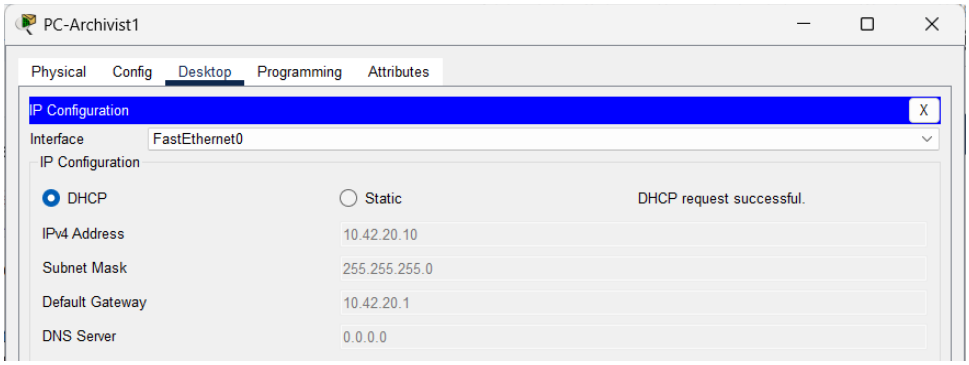


Figure 9: T2: PC-Archivist1 receives 10.42.20.10, gateway 10.42.20.1.

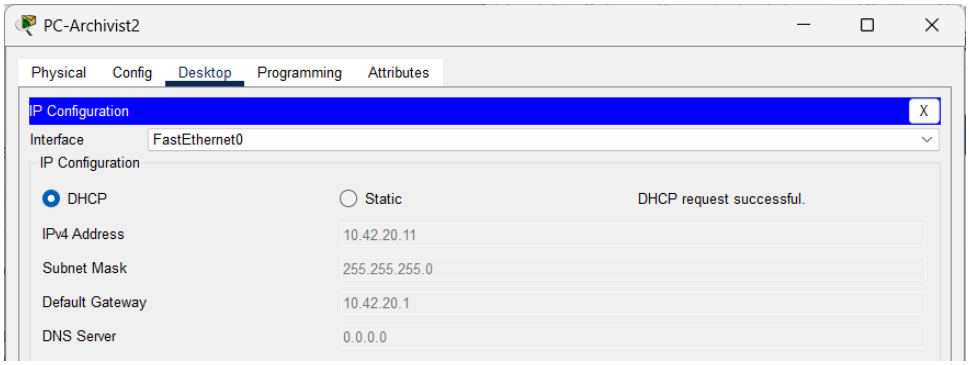


Figure 10: T2: PC-Archivist2 receives 10.42.20.11, gateway 10.42.20.1.

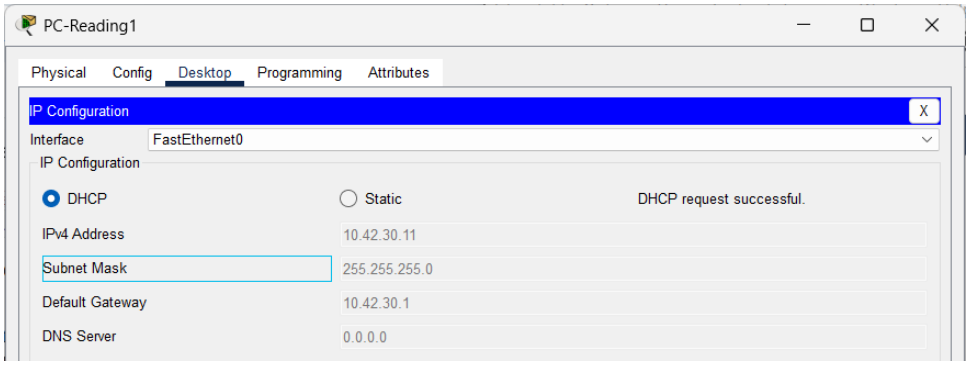


Figure 11: T3: PC-Reading1 receives 10.42.30.11, gateway 10.42.30.1.

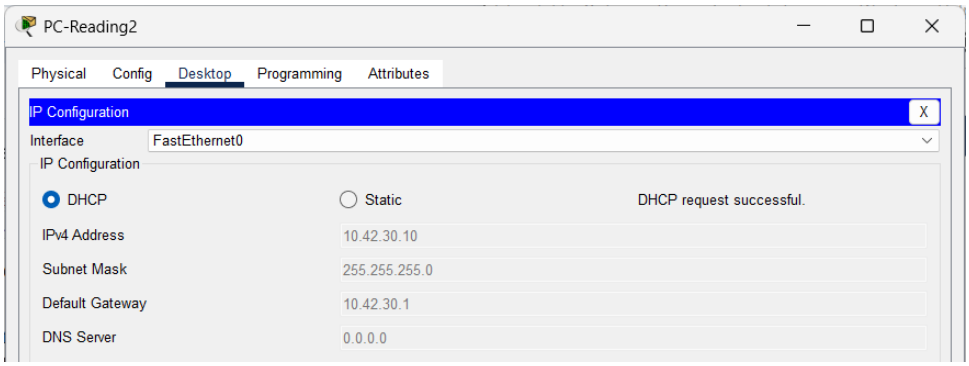


Figure 12: T3: PC-Reading2 receives 10.42.30.10, gateway 10.42.30.1.

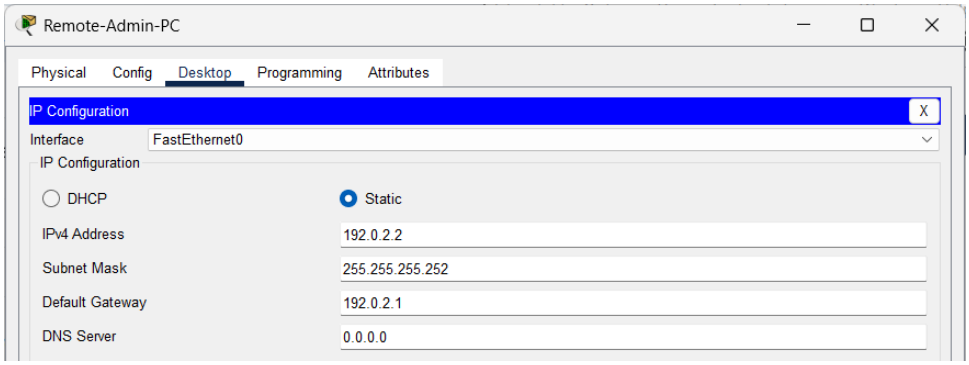


Figure 13: Remote-Admin-PC uses a static address: 192.0.2.2, mask 255.255.255.252, gateway 192.0.2.1.

6.3 Connectivity (T4, T5)

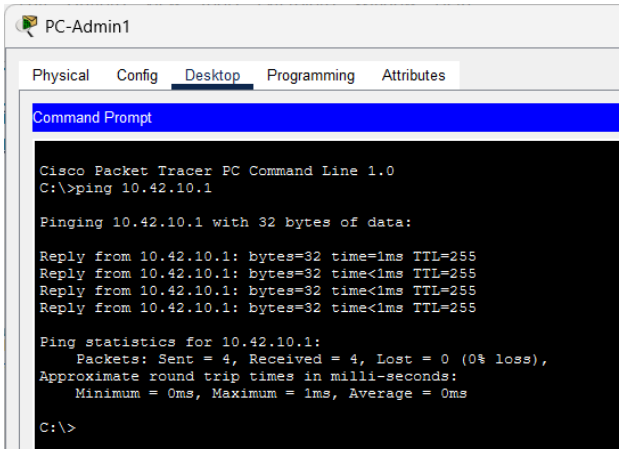


Figure 14: T4: PC-Admin1 pings 10.42.10.1 (4 of 4).

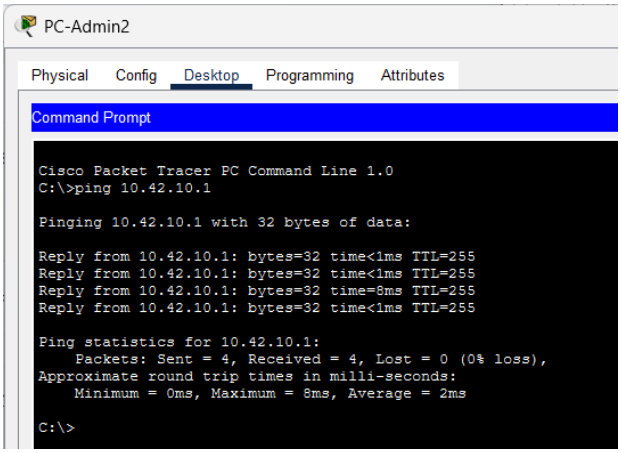


Figure 15: T4: PC-Admin2 pings 10.42.10.1 (4 of 4).

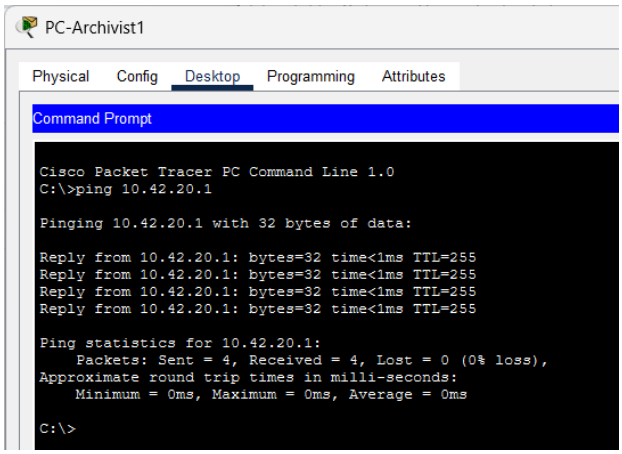


Figure 16: T4: PC-Archivist1 pings 10.42.20.1 (4 of 4).

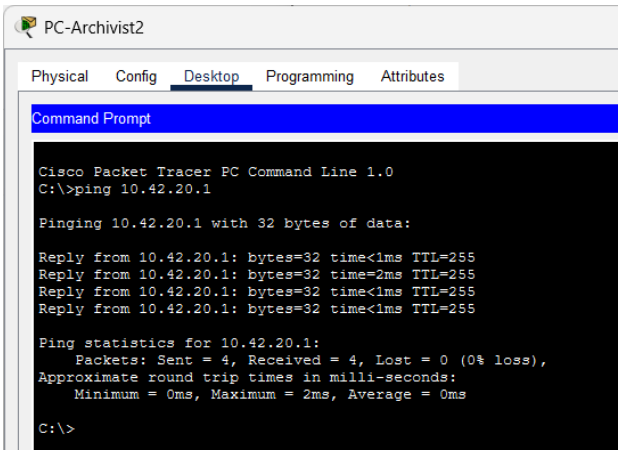


Figure 17: T4: PC-Archivist2 pings 10.42.20.1 (4 of 4).

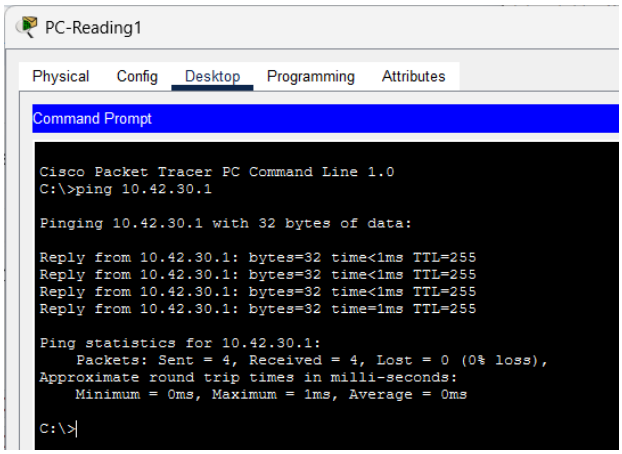


Figure 18: T4: PC-Reading1 pings 10.42.30.1 (4 of 4).

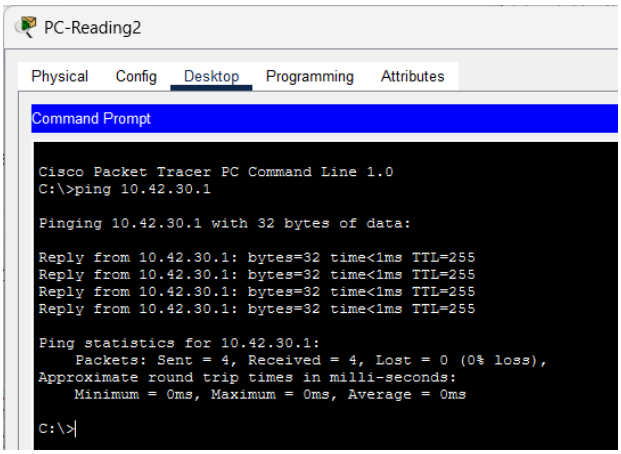


Figure 19: T4: PC-Reading2 pings 10.42.30.1 (4 of 4).

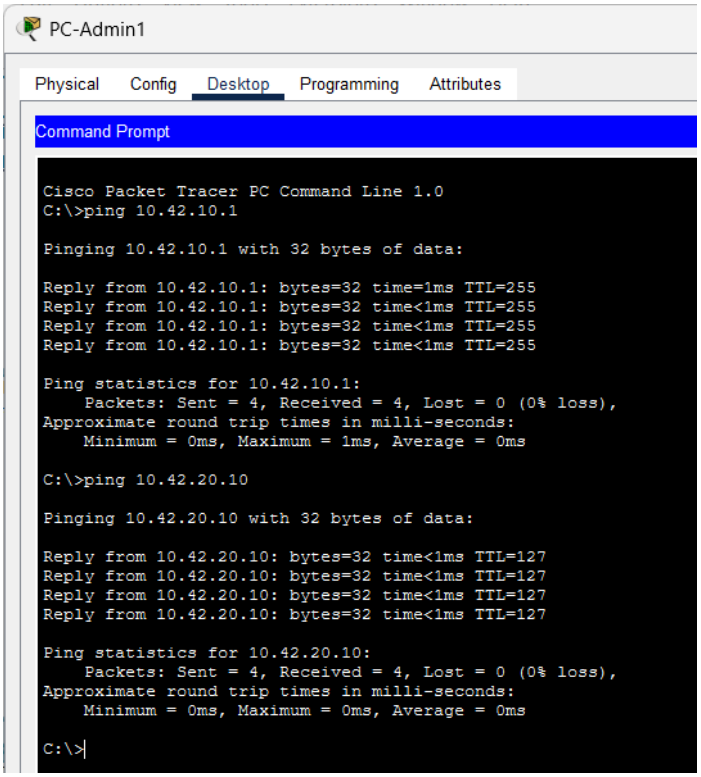


Figure 20: T5: PC-Admin1 pings its gateway, then PC-Archivist1 (10.42.20.10) in another VLAN: 4 of 4, TTL 127.

6.4 Switching and redundancy (T6, T7)

```
CORE-L3SW#enable
CORE-L3SW#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------

1      Po1(SU)           -      Gig1/0/23(P) Gig1/0/24(P)
CORE-L3SW#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Gig1/0/7, Gig1/0/8, Gig1/0/9, Gig1/0/10
                                                Gig1/0/11, Gig1/0/12, Gig1/0/13, Gig1/0/14
                                                Gig1/0/15, Gig1/0/16, Gig1/0/17, Gig1/0/18
                                                Gig1/0/19, Gig1/0/20, Gig1/0/21, Gig1/0/22
                                                Gig1/1/1, Gig1/1/2, Gig1/1/3, Gig1/1/4
10   Administration                   active    Gig1/0/2, Gig1/0/3
20   Archives-Research                active    Gig1/0/4, Gig1/0/5
30   Reading-Room                     active
40   Servers                          active    Gig1/0/1
99   Management                       active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
CORE-L3SW#
```

Figure 21: T6: CORE-L3SW shows Po1(SU) with Gi1/0/23(P) and Gi1/0/24(P), and the VLAN table with the host ports.

```
ACC-READSW#enable
ACC-READSW#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------

1      Po1(SU)           -      Fa0/23(P) Fa0/24(P)
ACC-READSW#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Gig0/1, Gig0/2
30   Reading-Room                     active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
99   Management                       active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
ACC-READSW#
```

Figure 22: T6: ACC-READSW shows Po1(SU) with Fa0/23(P) and Fa0/24(P); VLAN 30 holds Fa0/1 to Fa0/4.

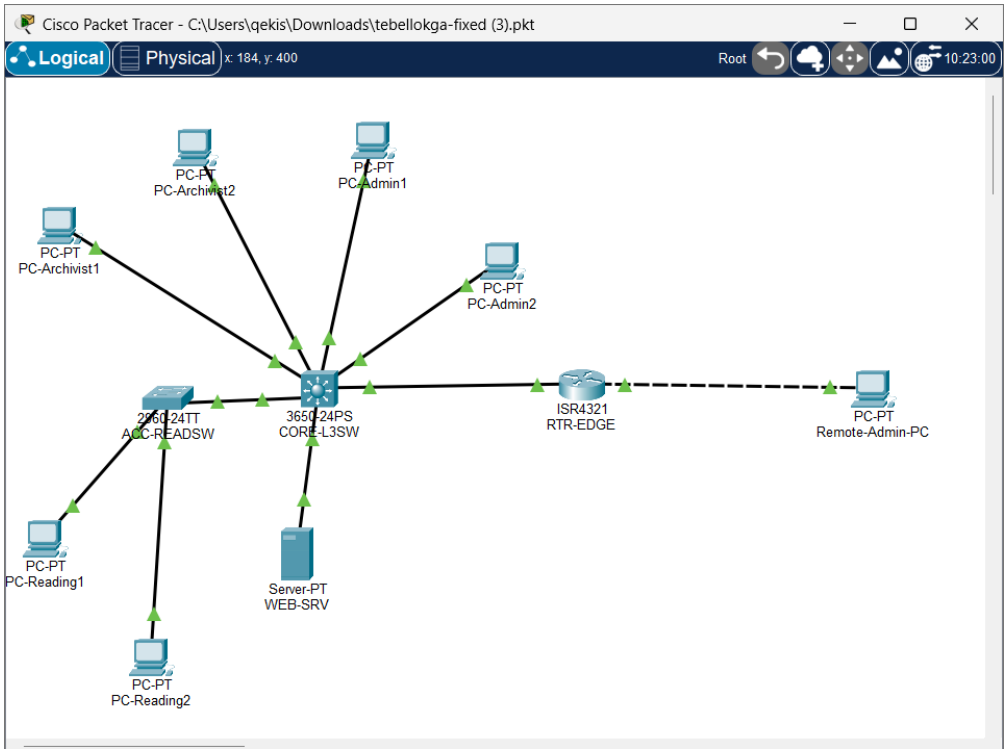


Figure 23: T7: one core-to-access cable removed; a single link remains between ACC-READSW and CORE-L3SW.

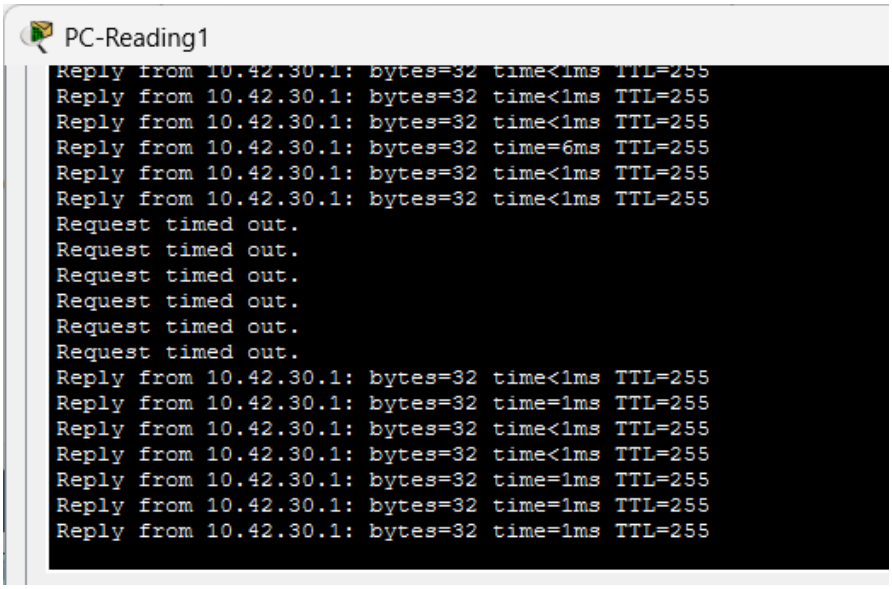


Figure 24: T7: continuous ping from PC-Reading1 to 10.42.30.1 loses 6 packets while the cable is removed, then replies resume.

```
Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------

1      Po1(SU)           -      Gig1/0/23(P) Gig1/0/24(D)
CORE-L3SW#
%LINK-5-CHANGED: Interface GigabitEthernet1/0/24, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/24, changed state to up

%LINK-3-UPDOWN: Interface GigabitEthernet1/0/23, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/23, changed state to down

%LINK-5-CHANGED: Interface GigabitEthernet1/0/23, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/23, changed state to up
show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------

1      Po1(SU)           -      Gig1/0/23(P) Gig1/0/24(P)
CORE-L3SW#
```

Figure 25: T7: CORE-L3SW shows Gi1/0/24(D) with Po1 still up, then both ports (P) after the cable is reconnected.

6.5 Assigned feature: restricted web server (T8 to T13)

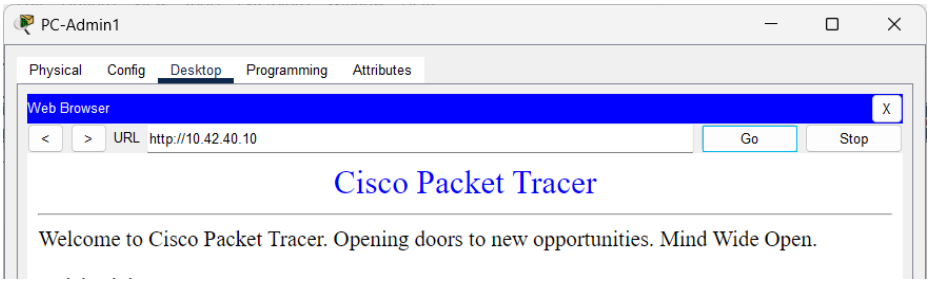


Figure 26: T8: PC-Admin1 loads the web page at http://10.42.40.10.

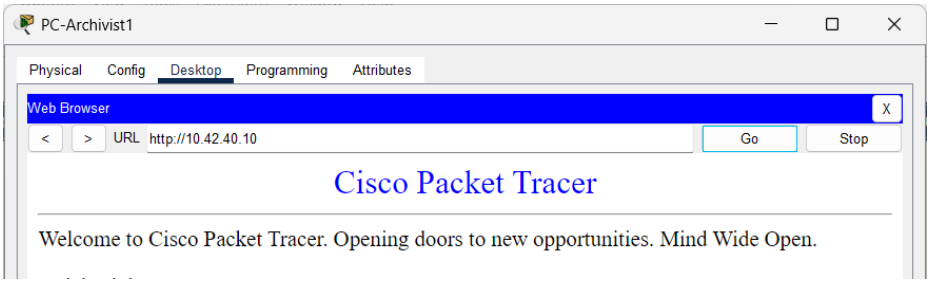


Figure 27: T9: PC-Archivist1 loads the web page at http://10.42.40.10.

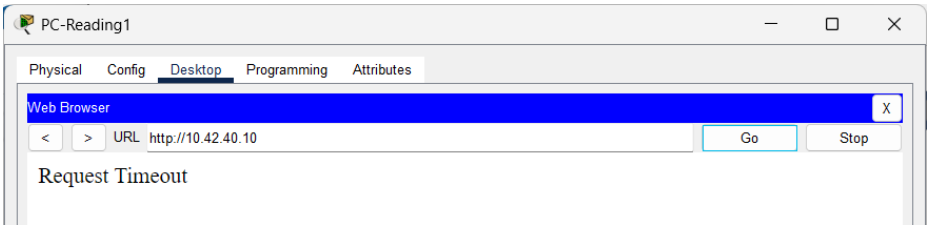


Figure 28: T10: PC-Reading1 cannot load the page (Request Timeout).

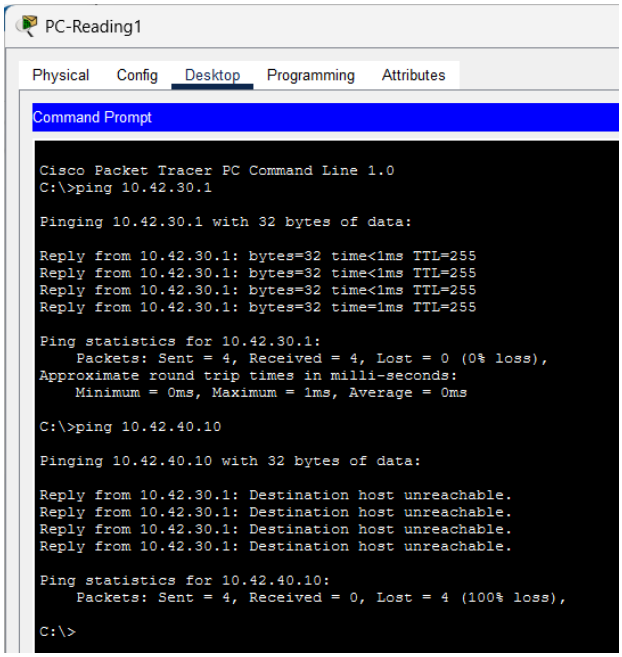


Figure 29: T11: PC-Reading1 reaches its gateway, but the ping to 10.42.40.10 gets "Destination host unreachable" (4 of 4 lost).

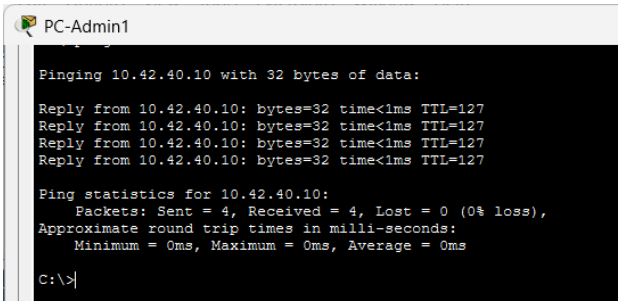


Figure 30: T12: PC-Admin1 pings the web server 10.42.40.10: 4 of 4 replies.

```
CORE-L3SW>enable
CORE-L3SW#enable
CORE-L3SW#show access-lists
Extended IP access list WEB-ACCESS
    10 permit tcp 10.42.10.0 0.0.0.255 host 10.42.40.10 eq www (5 match(es))
    20 permit tcp 10.42.20.0 0.0.0.255 host 10.42.40.10 eq www (5 match(es))
    30 permit icmp 10.42.10.0 0.0.0.255 10.42.40.0 0.0.0.255 (8 match(es))
    40 permit icmp 10.42.20.0 0.0.0.255 10.42.40.0 0.0.0.255
    50 deny ip any 10.42.40.0 0.0.0.255 (21 match(es))
    60 permit ip any any
Extended IP access list READING-ISOLATE
    10 deny ip 10.42.30.0 0.0.0.255 10.42.10.0 0.0.0.255 (4 match(es))
    20 deny ip 10.42.30.0 0.0.0.255 10.42.20.0 0.0.0.255
    30 deny ip 10.42.30.0 0.0.0.255 10.42.40.0 0.0.0.255 (16 match(es))
    40 deny ip 10.42.30.0 0.0.0.255 10.42.99.0 0.0.0.255 (16 match(es))
    50 permit ip any any (309 match(es))
Standard IP access list MGMT-SSH
    10 permit host 192.0.2.2 (4 match(es))
    20 permit 10.42.10.0 0.0.0.255 (6 match(es))
    30 permit 10.42.99.0 0.0.0.255
```

Figure 31: T13: show access-lists on CORE-L3SW after the tests, with match counters on WEB-ACCESS, READING-ISOLATE and MGMT-SSH.

Lines without a match count in Figure 31 (for example the ICMP permit for VLAN 20 and the denies from VLAN 30 to VLAN 20) were not exercised by a dedicated test. They follow the same pattern as the tested VLAN 10 lines and were checked in the configuration.

6.6 Isolation and the edge (T14 to T17)

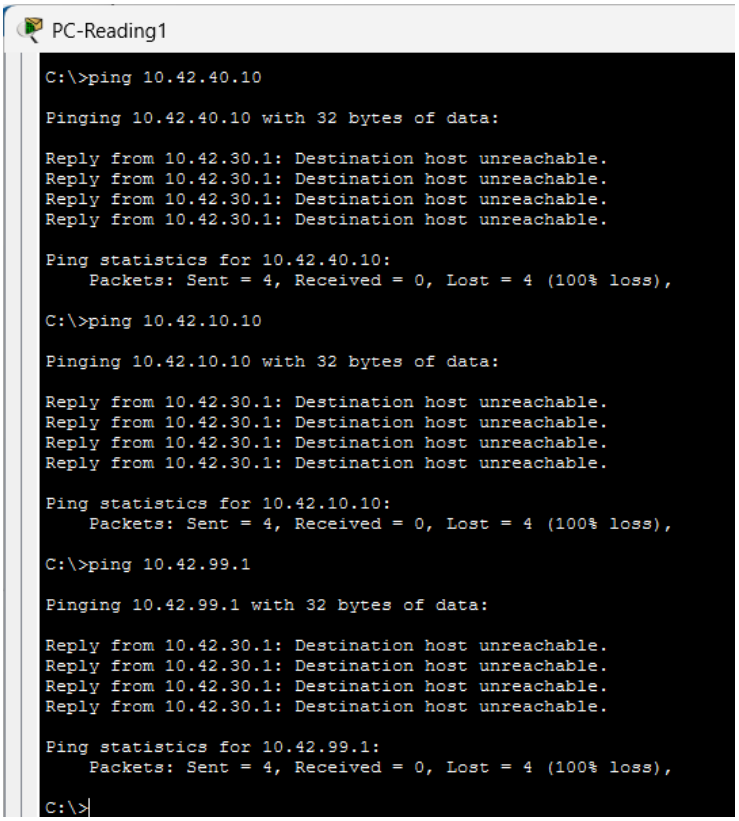


Figure 32: T14: PC-Reading1 cannot reach the server, PC-Admin1 (10.42.10.10) or the management SVI (10.42.99.1).

```
RTR-EDGE>ping 10.42.0.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.42.0.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

RTR-EDGE>
```

Figure 33: T15: RTR-EDGE pings the core transit address 10.42.0.2 (5 of 5).

```
CORE-L3SW>ping 10.42.0.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.42.0.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

CORE-L3SW>
```

Figure 34: T15: CORE-L3SW pings the router transit address 10.42.0.1 (5 of 5).

```
RTR-EDGE>ping 192.0.2.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.0.2.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/12/30 ms

RTR-EDGE>ping 10.42.10.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.42.10.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/7/38 ms

RTR-EDGE>ping 10.42.40.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.42.40.10, timeout is 2 seconds:
UUUUU
Success rate is 0 percent (0/5)

RTR-EDGE>
```

Figure 35: T15 and T17: RTR-EDGE reaches 192.0.2.1 and 10.42.10.1 (5 of 5), but its ping to the server 10.42.40.10 fails (UUUUU).

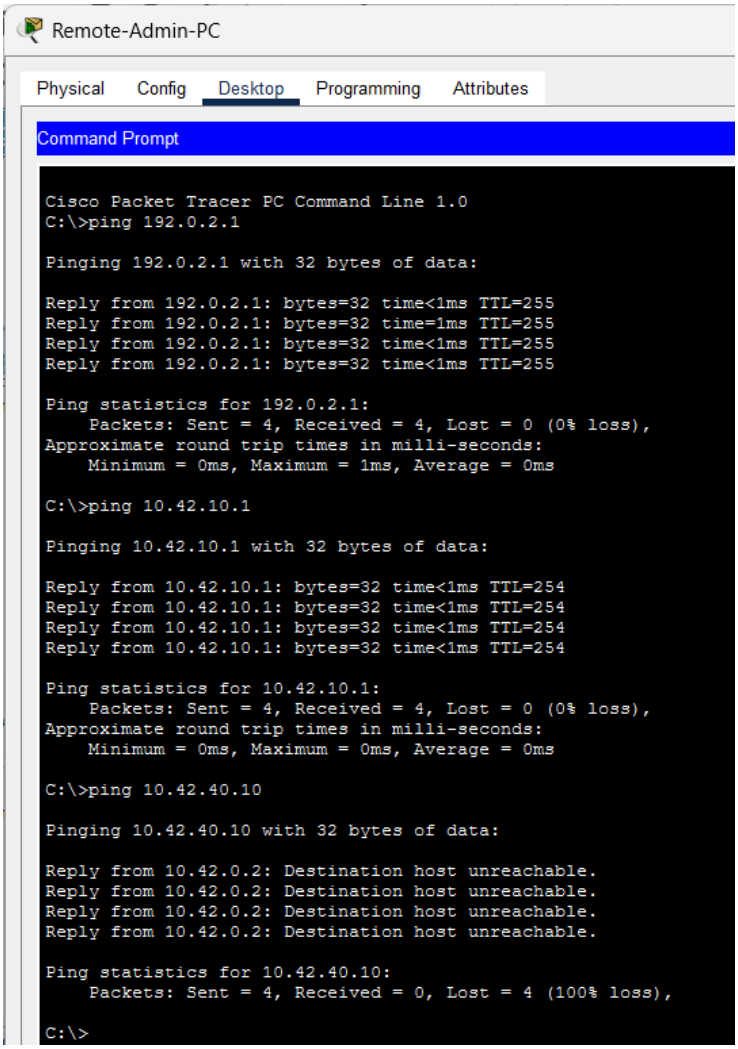


Figure 36: T16 and T17: Remote-Admin-PC reaches 192.0.2.1 and 10.42.10.1, but its ping to 10.42.40.10 is refused by the core (10.42.0.2).

6.7 Remote management (T18 to T20)

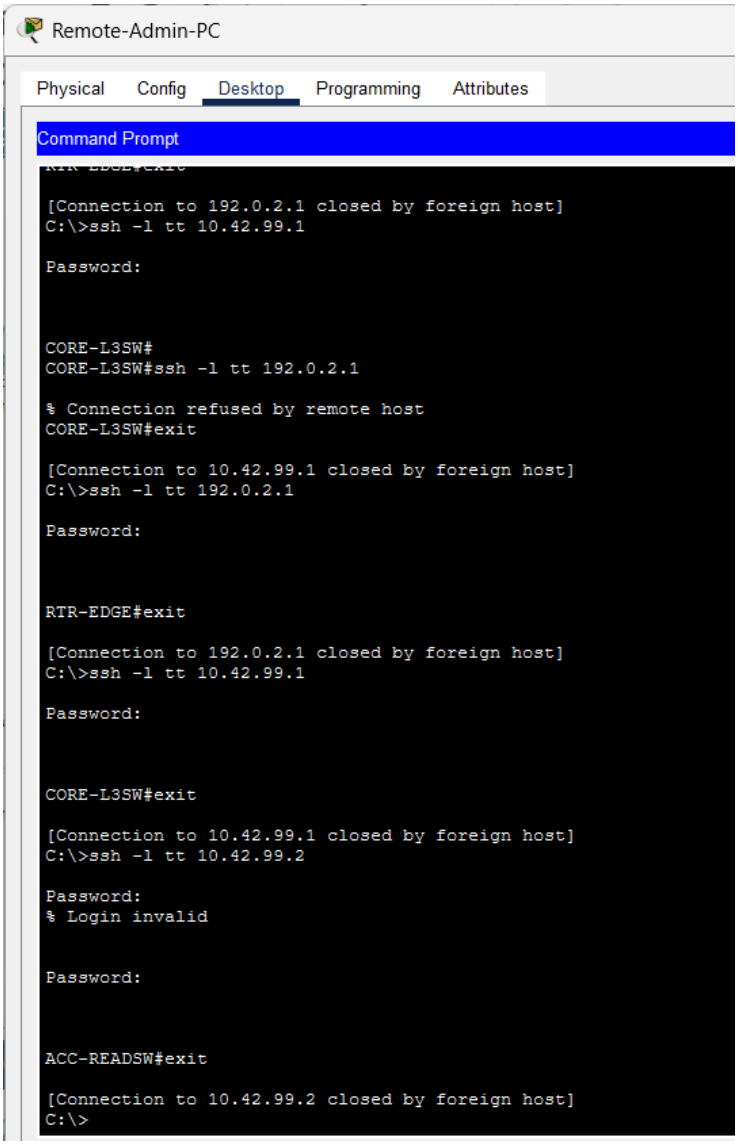


Figure 37: T18: SSH from Remote-Admin-PC to 192.0.2.1, 10.42.99.1 and 10.42.99.2 reaches each privileged prompt. From CORE-L3SW, SSH to the router is refused. (One mistyped password attempt on ACC-READSW is visible before the successful login.)

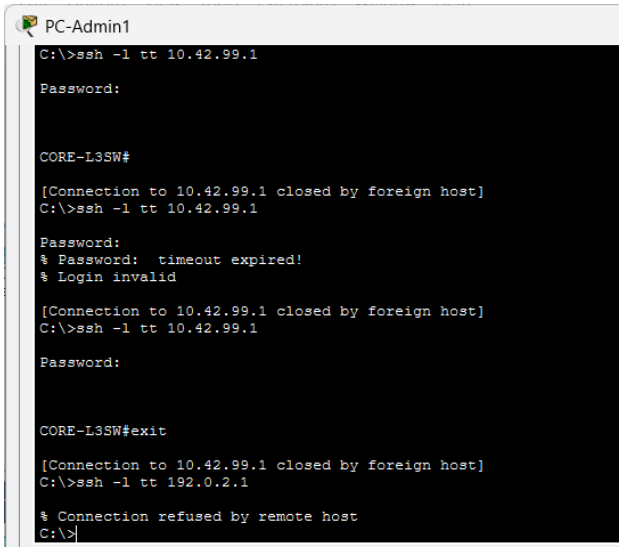


Figure 38: T19: PC-Admin1 can SSH to the core (allowed by MGMT-SSH) but is refused by the router.

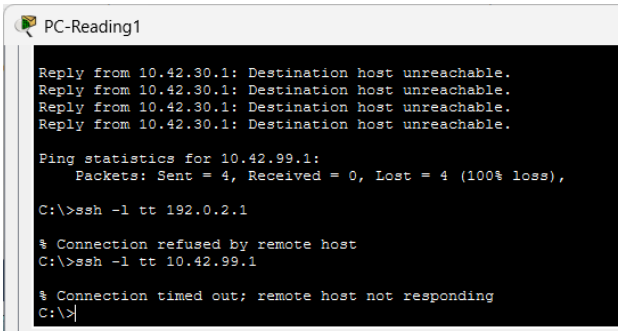


Figure 39: T19: PC-Reading1 is refused by the router and times out towards the management address.

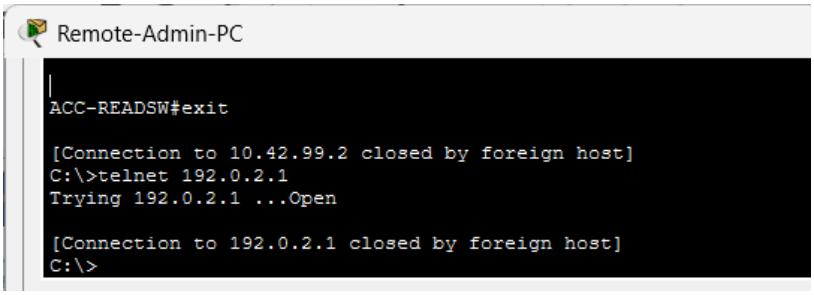


Figure 40: T20: telnet to 192.0.2.1 opens, then the router closes the connection at once with no login prompt.

6.8 Saved configurations (T21)

```
CORE-L3SW#write memory
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
CORE-L3SW#show startup-config
Using 3702 bytes
!
version 16.3.2
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname CORE-L3SW
!
!
no profinet
!
!
ip dhcp excluded-address 10.42.10.1 10.42.10.9
ip dhcp excluded-address 10.42.20.1 10.42.20.9
ip dhcp excluded-address 10.42.30.1 10.42.30.9
!
ip dhcp pool ADMIN-POOL
 network 10.42.10.0 255.255.255.0
 default-router 10.42.10.1
ip dhcp pool ARCHIVE-POOL
 network 10.42.20.0 255.255.255.0
 --More--
```

Figure 41: T21: CORE-L3SW write memory [OK] and show startup-config (top of the saved configuration).

```
RTR-EDGE>enable
RTR-EDGE#write memory
Building configuration...
[OK]
RTR-EDGE#show startup-config
Using 787 bytes
!
version 15.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname RTR-EDGE
!
!
!
!
!
!
!
ip cef
no ipv6 cef
!
!
!
username tt privilege 15 secret 5 $1$mERr$7J4tH5Cv.YVmTBEGC5OXU1
 --More--
```

Figure 42: T21: RTR-EDGE write memory [OK] and show startup-config (top of the saved configuration).

```
ACC-READSW#enable
ACC-READSW#write memory
Building configuration...
[OK]
ACC-READSW#show startup-config
Using 1839 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname ACC-READSW
!
!
!
ip ssh version 2
ip domain-name dharc.local
!
username tt secret 5 $1$mERr$7J4tH5Cv.YVmTBEGC5OXU1
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface Port-channel1
 --More--
```

Figure 43: T21: ACC-READSW write memory [OK] and show startup-config (top of the saved configuration).

7. Notes and limitations

- **Failover is short, not zero:** removing one EtherChannel cable cost about 6 pings (T7) while the bundle re-formed on the remaining link; connectivity then recovered without manual action. The design survives a cable failure but does not hide it completely.
- **Telnet behaviour:** the router accepts the TCP connection and then closes it immediately because the vty lines allow SSH only (T20). There is no login prompt, so Telnet cannot be used, but Packet Tracer does not print a "refused" message.
- **Who may manage the switches:** MGMT-SSH deliberately allows the Administration VLAN as well as the off-site administrator, so on-site administrators can SSH to the core (Figure 38). The router is stricter and accepts only the off-site administrator.
- **Credentials:** SSH logins use local user tt with a password stored as a secret on each device. The enable secret is not set and the console line has no password, which is acceptable inside a simulation; a production network would set both. Passwords are not recorded in this report or in the repository.

- **Static design choices:** static routing and a static EtherChannel (mode on) were chosen for simplicity. LACP would negotiate the bundle and detect a misconfigured partner, and a dynamic routing protocol would be considered if more routers were added.
- **Single points of failure:** there is one edge router and one core switch, so those remain single points of failure. The EtherChannel removes the single point of failure only for the core-to-access link.

8. Contents of the submission

Item	Purpose
Packet Tracer project (.pkt)	Working implementation, saved after testing with RSA keys and local users in place
This report	Implementation description, assigned feature, fault audit and testing evidence
GitHub repository dharc-archive-network	Portfolio of evidence: configurations, screenshots and change history

Appendix A: Configuration audit

Before the live tests, a script decoded the project file and checked 52 items of stored configuration and cabling: VLAN membership, cabling, EtherChannel members, routes, DHCP pools, ACL decisions for the key flows, SSH settings and saved configurations. It is a configuration check only; the live tests in Section 6 are the proof that traffic behaves correctly.

Run	Checks passed	Checks failed
Original file (before corrections)	36 of 52	16
Corrected file (before live testing)	49 of 52	3 (RSA keys, generated live in Figures 4 to 6)

Appendix B: Running configurations

Configurations of the corrected project before live testing; the local user password hash is not shown. RSA keys and the local passwords were added during testing. Unused interfaces without configuration are omitted.

RTR-EDGE

```
version 15.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
hostname RTR-EDGE
ip cef
no ipv6 cef
username tt privilege 15 secret 5 <hash not shown>
ip ssh version 2
ip domain-name dharc.local
spanning-tree mode pvst
interface GigabitEthernet0/0/0
 ip address 192.0.2.1 255.255.255.252
 duplex auto
 speed auto
interface GigabitEthernet0/0/1
 ip address 10.42.0.1 255.255.255.252
 duplex auto
 speed auto
ip classless
ip route 10.42.0.0 255.255.0.0 10.42.0.2
ip flow-export version 9
access-list 10 permit host 192.0.2.2
line con 0
line aux 0
line vty 0 4
 access-class 10 in
 login local
 transport input ssh
end
```

CORE-L3SW

```
version 16.3.2
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
hostname CORE-L3SW
no profinet
ip dhcp excluded-address 10.42.10.1 10.42.10.9
ip dhcp excluded-address 10.42.20.1 10.42.20.9
```

```
ip dhcp excluded-address 10.42.30.1 10.42.30.9
ip dhcp pool ADMIN-POOL
 network 10.42.10.0 255.255.255.0
 default-router 10.42.10.1
ip dhcp pool ARCHIVE-POOL
 network 10.42.20.0 255.255.255.0
 default-router 10.42.20.1
ip dhcp pool READING-POOL
 network 10.42.30.0 255.255.255.0
 default-router 10.42.30.1
no ip cef
ip routing
no ipv6 cef
username tt privilege 15 secret 5 <hash not shown>
ip ssh version 2
ip domain-name dharc.local
spanning-tree mode pvst
interface Port-channel1
 switchport mode trunk
interface GigabitEthernet1/0/1
 switchport access vlan 40
 switchport mode access
interface GigabitEthernet1/0/2
 switchport access vlan 10
 switchport mode access
interface GigabitEthernet1/0/3
 switchport access vlan 10
 switchport mode access
interface GigabitEthernet1/0/4
 switchport access vlan 20
 switchport mode access
interface GigabitEthernet1/0/5
 switchport access vlan 20
 switchport mode access
interface GigabitEthernet1/0/6
 no switchport
 ip address 10.42.0.2 255.255.255.252
interface GigabitEthernet1/0/23
 switchport mode trunk
 channel-group 1 mode on
interface GigabitEthernet1/0/24
 switchport mode trunk
 channel-group 1 mode on
interface Vlan1
 no ip address
 shutdown
interface Vlan10
 mac-address 00d0.582c.5c01
 ip address 10.42.10.1 255.255.255.0
interface Vlan20
 mac-address 00d0.582c.5c02
 ip address 10.42.20.1 255.255.255.0
interface Vlan30
 mac-address 00d0.582c.5c03
 ip address 10.42.30.1 255.255.255.0
 ip access-group READING-ISOLATE in
interface Vlan40
 mac-address 00d0.582c.5c04
 ip address 10.42.40.1 255.255.255.0
 ip access-group WEB-ACCESS out
interface Vlan99
 mac-address 00d0.582c.5c05
 ip address 10.42.99.1 255.255.255.0
ip classless
ip route 0.0.0.0 0.0.0.0 10.42.0.1
ip flow-export version 9
ip access-list extended WEB-ACCESS
 permit tcp 10.42.10.0 0.0.0.255 host 10.42.40.10 eq www
 permit tcp 10.42.20.0 0.0.0.255 host 10.42.40.10 eq www
```

```
 permit icmp 10.42.10.0 0.0.0.255 10.42.40.0 0.0.0.255
 permit icmp 10.42.20.0 0.0.0.255 10.42.40.0 0.0.0.255
 deny ip any 10.42.40.0 0.0.0.255
 permit ip any any
ip access-list extended READING-ISOLATE
 deny ip 10.42.30.0 0.0.0.255 10.42.10.0 0.0.0.255
 deny ip 10.42.30.0 0.0.0.255 10.42.20.0 0.0.0.255
 deny ip 10.42.30.0 0.0.0.255 10.42.40.0 0.0.0.255
 deny ip 10.42.30.0 0.0.0.255 10.42.99.0 0.0.0.255
 permit ip any any
ip access-list standard MGMT-SSH
 permit host 192.0.2.2
 permit 10.42.10.0 0.0.0.255
 permit 10.42.99.0 0.0.0.255
line con 0
line aux 0
line vty 0 4
 access-class MGMT-SSH in
 login local
 transport input ssh
line vty 5 15
 access-class MGMT-SSH in
 login local
 transport input ssh
end
```

ACC-READSW

```
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
hostname ACC-READSW
ip ssh version 2
ip domain-name dharc.local
username tt privilege 15 secret 5 <hash not shown>
spanning-tree mode pvst
spanning-tree extend system-id
interface Port-channel1
 switchport mode trunk
interface FastEthernet0/1
 switchport access vlan 30
 switchport mode access
interface FastEthernet0/2
 switchport access vlan 30
 switchport mode access
interface FastEthernet0/3
 switchport access vlan 30
 switchport mode access
interface FastEthernet0/4
 switchport access vlan 30
 switchport mode access
interface FastEthernet0/23
 switchport mode trunk
 channel-group 1 mode on
interface FastEthernet0/24
 switchport mode trunk
 channel-group 1 mode on
interface Vlan1
 no ip address
 shutdown
interface Vlan99
 ip address 10.42.99.2 255.255.255.0
ip default-gateway 10.42.99.1
ip access-list standard MGMT-SSH
 permit host 192.0.2.2
 permit 10.42.10.0 0.0.0.255
```

```
 permit 10.42.99.0 0.0.0.255
line con 0
line vty 0 4
 access-class MGMT-SSH in
 login local
 transport input ssh
line vty 5 15
 access-class MGMT-SSH in
 login local
 transport input ssh
end
```